

Plotting and honest charts

Build a chart in matplotlib, choose its type from your question, and set the axes so the picture matches the numbers.

THE QUESTION

Which chart answers my question, and how do I draw it so the picture says what the numbers say?

THE INTUITION

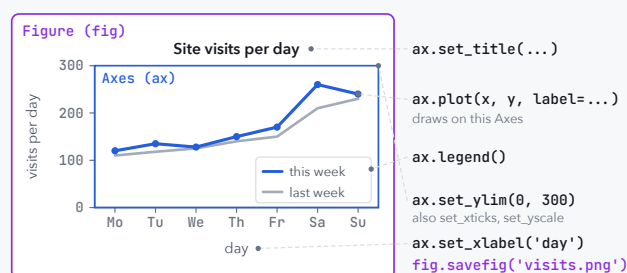
A matplotlib Figure holds one or more Axes, and you draw on an Axes with plot, scatter, bar, hist or boxplot. Readers judge lengths, positions and areas, so the baseline, the scale and the bin width decide what they see.

KEY TAKEAWAY

Choose the chart from the question. Start bars at zero, label axes with units, and show the spread as well as the average.

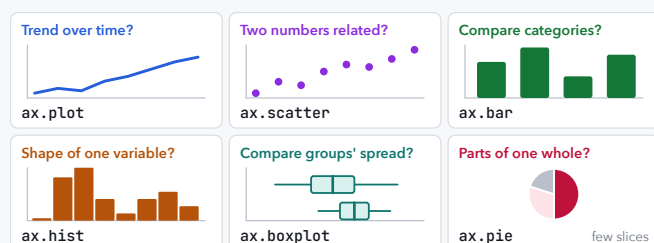
question → fig, ax → plot · scatter · bar → hist · boxplot → limits, scale → labels, units → savefig

1 A Figure holds Axes; you draw on the Axes



fig, ax = plt.subplots() makes a Figure (the canvas that savefig writes) holding one Axes (one plot with its own axes). Call methods on ax. fig, axes = plt.subplots(2, 2) gives a 2 by 2 array of Axes; draw on axes[0, 1]. plt.plot draws on the current Axes, which gets confusing with several.

2 The question picks the chart



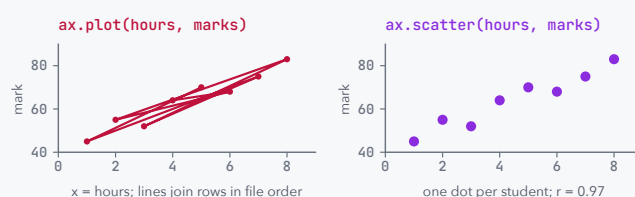
Line: a quantity over an ordered x, usually time. Scatter: two numbers measured on the same items. Bar: one number per category. Histogram: how one variable's values spread; box plots set groups side by side. A pie shows shares of one whole; past a few slices, bars compare better.

3 The code behind panel 1

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(5, 3))
ax.plot(days, now, "o-", label="this week")
ax.plot(days, last, label="last week")
ax.set_title("Site visits per day")
ax.set_xlabel("day")
ax.set_ylabel("visits per day")
ax.set_ylim(0, 300); ax.legend()
fig.savefig("visits.png", dpi=300)
```

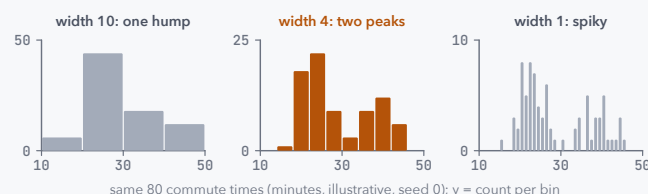
days holds the 7 day names; now and last hold the counts. pandas: df.plot(x="day") returns an Axes too, so the same set_ calls work. Save PDF or SVG for print.

4 Join points only when x has an order



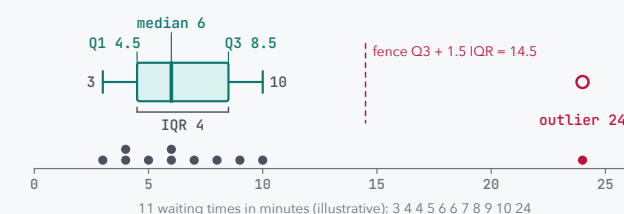
Eight students' study hours and marks (illustrative), stored in the order they were recorded. ax.plot joins consecutive rows, so it draws a scribble. ax.scatter, or ax.plot(x, y, "o"), gives one dot per student and shows the upward pattern. A y-axis from 40 is fine here: dots are read by position, not by length.

5 Bin width changes the story



ax.hist(x, bins=...) counts the values in each bin. Wide bins merge the two groups of commuters into one hump; 1-minute bins hold 0 to 8 values each, which is mostly noise. Try a few widths, keep the shape that stays, and say which width you used. bins=np.arange(10, 51, 4) fixes the edges; density=True makes the bar areas sum to 1.

6 A box plot shows quartiles and outliers

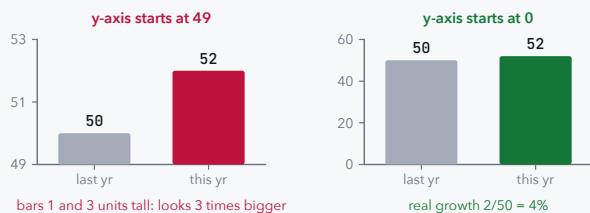


The box runs from Q1 (a quarter of the values lie below it) to Q3 (three quarters below); the line inside is the median. Here Q1 = 4.5, median 6, Q3 = 8.5, so the interquartile range IQR = 4. Whiskers reach the most extreme values within 1.5 IQR of the box, 3 and 10; 24 lies past the fence and is drawn as a point. ax.boxplot uses this rule by default.

Honest charts: baselines, scales and spread

Readers compare lengths, positions and areas. Set up the axes so those comparisons match the numbers.

7 Bars start at zero



A bar is read by its length. Sales grew from 50 to 52 (illustrative), 4%. On a y-axis starting at 49 the bars are 1 and 3 units tall, so the new bar looks 3 times bigger. matplotlib starts bars at 0 unless you call `set_ylim` yourself. Line and scatter charts may zoom in, since they are read by position.

8 A log scale shows ratios



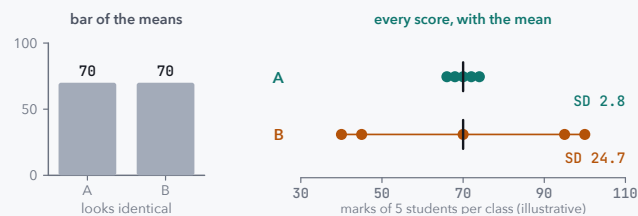
Users grow 10, 100, 1,000, 10,000 over four years (illustrative). On a linear axis the first steps (+90, +900) vanish near zero. `ax.set_yscale("log")` spaces powers of ten evenly, so steady x10 growth is a straight line. It suits data spanning several powers of ten and cannot show 0 or negatives. Tick labels stay in real units: 10, 100, 1,000.

9 An outlier drags the mean, not the median



Nine people earn 30,000 and one earns 300,000 (illustrative). The mean is $570,000 / 10 = 57,000$, more than 9 of the 10 earn. The median, the middle value once sorted, stays at 30,000; without the top earner the mean is 30,000 as well. For skewed data such as pay or waiting times, report the median, or both.

10 An average hides the spread



Class A scored 66, 68, 70, 72, 74 and class B 40, 45, 70, 95, 100 (illustrative). Both means are 70, so a bar chart of the means shows twins. The standard deviation (dividing by n , as `np.std` does) is 2.8 for A and 24.7 for B. Plot the values: dots, a box plot or a histogram per group. If a bar shows a mean, add error bars with `ax.errorbar` and say what they measure.

11 A chart should explain itself: title, axis labels with units, a legend



The same two weeks of visits (illustrative), bare and labelled. From the chart alone a reader should learn what was measured, in which unit, and which line is which. Use `set_xlabel` and `set_ylabel("visits per day")`, give each plot a `label=` and call `ax.legend()`. Tick labels people read (Mon, not 0), a title that says what is shown and a note on the data source finish it.

12 Pause and think

A line chart's y-axis starts at 60. Misleading? Not by itself: lines are read by position and slope. Show the axis values clearly. 1, 10, 100, 1000 on a log axis: how far apart? Evenly spaced, one step per factor of 10.

`bins=5` shows one peak, `bins=20` two. Which is right? Try widths in between. A peak that stays across sensible widths is real. Can a log axis show 0? No. $\log 0$ is minus infinity, so 0 has no place on the axis; matplotlib clips or drops it.

Mean pay 57,000, median 30,000. Which way is the skew? To the right: a few large values pull the mean up. When is a pie chart fine? A few slices of one whole. To compare slices with each other, bars work better.

question first, then chart · bars from zero · log scale for ratios · show the spread · label with units

Source: the course notebook on matplotlib (figures and axes, plot types, limits and ticks, scale transformations), nipunbatra.github.io/psdv-teaching, used for the list of topics. All examples and numbers are our own (illustrative) and computed by the sheet's figure script.